Experimental Setup and Observations

IEEE-paper support document for the grounded meeting-intelligence pipeline

Purpose. This report presents the experimental configuration, measurement protocol, recorded results, and engineering observations supporting the IEEE manuscript. It separates directly recorded validation evidence from manuscript values that still require their underlying datasets and scripts for independent reproduction.

Evidence status. The primary reproducibility result is a five-run forced re-analysis of one controlled navigation fixture. Transcription throughput is measured independently on one audio recording with 30 s, 60 s, and 120 s clips, each repeated three times.

Recorded artifact	Value
Consistency validation	Passed; 5 forced runs
Core output per run	2 actions, 1 decision, 1 question
Minimum core Jaccard overlap	1.00 for actions, decisions, questions
Transcription benchmark	9 timed runs; 30/60/120 s clips
Runtime platform	Windows 11; 16 logical CPUs; Python 3.12.10

I. Experimental Setup

A. System under evaluation

The evaluated system is a hybrid, evidence-grounded meeting analyzer. Audio/video is decoded with FFmpeg and transcribed locally by Faster-Whisper using CTranslate2. The transcript is preserved with ordered segments and timestamps. Classical NLP components provide preprocessing, named-entity recognition, and dialogue-act signals. Structured semantic extraction then produces actions, decisions, questions, follow-ups, topics, and summaries. Every core insight is validated against transcript evidence before it is deduplicated and persisted.

Stage	Implementation / control
Input	Audio/video upload; FFmpeg extraction and format validation
ASR	Faster-Whisper base; CTranslate2; INT8 CPU inference; VAD enabled
Linguistic processing	spaCy preprocessing, NER, TF-IDF features, dialogue-act classification
Semantic extraction	Groq openai/gpt-oss-120b; strict JSON Schema; temperature 0; top-p 1
Evidence control	Verbatim quote matching to source segments; unsupported evidence rejected
State control	SHA-256 manifests, stable source-anchored IDs, deterministic reconciliation
Output	Canonical report, dashboard data, and shareable PDF export

B. Runtime and model configuration

Parameter	Configuration
Operating system / CPU	Windows 11; AMD64; 16 logical CPUs
Python and packages	Python 3.12.10; faster-whisper 1.2.1; CTranslate2 4.8.2
ASR model	Systran/faster-whisper-base; recorded revision ebe41f70...
ASR decoding	Beam size 5; VAD filter true; minimum silence 500 ms; language auto
LLM	Groq API, openai/gpt-oss-120b; strict json_schema
LLM controls	Temperature 0; top-p 1; reasoning effort low; max output 8192 tokens
Determinism	SHA-256 request-content seed policy; ordered segment chunks; deterministic merge

C. Datasets and test fixtures

The manuscript describes AMI/ICSI corpora, an internal engineering-meeting corpus, a controlled navigation fixture, and multilingual tests. The directly auditable artifacts used for this report are the controlled navigation fixture for semantic consistency and one independent recording for ASR throughput. The fixture contains seven timestamped segments and was held constant across all five forced analyses.

D. Measurement protocol

For semantic stability, the same canonical input, prompt version, schema, analysis version, and provider configuration were submitted five times with force=true, bypassing ordinary cached-result reuse. Outputs were measured at generated, parsed, validated, evidence-matched, deduplicated, and persisted stages. For ASR throughput, warm transcription runs were timed after model loading; decode, VAD, language detection, lazy segment consumption, and output construction were included, while model loading, NLP, and persistence were excluded.

II. Metrics and Recorded Results

A. Evaluation metrics

Metric	Definition and interpretation
Jaccard overlap	J(A,B) = |A intersection B| / |A union B|, using stable item identity and source evidence.
Stage retention	Items surviving each validation boundary divided by items entering that boundary.
Real-time factor	Transcription time divided by clip duration; values below 1 indicate faster-than-real-time processing.
Run variance	Mean, standard deviation, minimum, and maximum of three timed runs per clip length.
Grounding status	An item is evidence-matched only when its supporting quote maps to the original transcript segment.

B. Five-run semantic consistency

Category	Run 1	Run 2	Run 3	Run 4	Run 5	Min J
Action items	2	2	2	2	2	1.00
Decisions	1	1	1	1	1	1.00
Questions	1	1	1	1	1	1.00
Follow-ups	0	0	0	0	0	N/A

The five forced runs returned identical core counts: 2 action items, 1 decision, and 1 question. The empty follow-up category is not a recall measurement because the fixture contains no separate follow-up.

C. Pipeline retention audit

Category	Generated	Parsed	Validated	Evidence matched	Deduplicated
Action items	2	2	2	2	2
Decisions	1	1	1	1	1
Questions	1	1	1	1	1
Follow-ups	0	0	0	0	0

For the recorded fixture, every generated core item survived all local validation boundaries. Across five runs this corresponds to 20 of 20 category-run core items remaining evidence matched. This is a fixture-level observation, not a universal hallucination-rate estimate.

D. Local transcription throughput

Clip (s)	Mean (s)	SD (s)	Min (s)	Max (s)	Mean RTF
30	2.315	0.483	1.757	2.614	0.077
60	5.715	0.563	5.065	6.041	0.095
120	11.562	0.537	11.063	12.130	0.096

Mean real-time factors remained below 0.10 for all tested durations. The 120-second clip averaged 11.562 seconds of transcription time (RTF 0.096), indicating faster-than-real-time local ASR under the recorded CPU configuration. The benchmark does not include full upload, NLP, LLM, or persistence latency.

III. Detailed Observations

1. **Core semantic identities were stable.** The same source-anchored action items, decision, and question were reproduced across all five forced analyses. Stability therefore held at the level most relevant to report persistence: category, semantic identity, and source evidence.
2. **Local reconciliation removed provider-order dependence.** The candidate and persisted counts were identical in every run. Stable IDs, normalized matching, evidence checks, and deterministic merging prevent a later response from silently replacing a valid earlier report.
3. **Evidence validation was an effective boundary on this fixture.** No core item was removed between generation and evidence matching. The result is consistent with the system's design: supporting quotes must map to transcript segments, and invalid or unsupported evidence is not persisted as a successful insight.
4. **Descriptive topics are less stable than core outcomes.** The recorded provenance shows topic overlap varying from 0.00 to 1.00 across runs, while actions, decisions, and questions remained at 1.00. This suggests that topic labels should be treated as descriptive context rather than as audit-grade identifiers.
5. **Local ASR throughput was consistent across scale.** Mean RTF was 0.077, 0.095, and 0.096 for 30-, 60-, and 120-second clips. The near-flat ratio after the shortest clip indicates that fixed overhead was amortized, but the result is hardware- and model-specific.
6. **Reproducibility metadata is part of the result.** The validation manifest records input, configuration, schema, system, and model hashes. These fields allow a future run to distinguish a changed input or pipeline from genuine model variation.

IV. Reproducibility and Limitations

Semantic runs used analysis version **meeting-intelligence-v3**, prompt version **exhaustive-evidence-v1**, model **openai/gpt-oss-120b**, temperature **0**, top-p **1**, strict JSON Schema output, and an input hash beginning `1615c5fecfbb465c...`.

The controlled fixture is short, English, and single-purpose. It does not establish accuracy across speakers, domains, languages, noisy recordings, or long meetings. It also does not independently reproduce manuscript-only claims such as AMI/ICSI classification scores, ROUGE values, or an unconstrained-LLM baseline. Those claims should be retained in the IEEE paper only after the corresponding datasets, predictions, references, and measurement scripts are archived.

Source artifacts: data/validation/consistency/results.json, data/validation/consistency/consistency_pairs.csv, and research_evaluation/data/transcription_summary.csv.